Regular thin covers with disconnected face intersections

Human Compression project

Author edition; pending independent review

Abstract

We express the cyclic voltage family of Leemans and Toledo as a fiber product with a dihedral group. Two independent binary modules describe its endpoint parabolics. Their intersections then follow from the same projection and kernel argument used for mixes. Taking a canonical double cover gives the faithful, thin, nonpolytopal family; adjoining commuting colors gives every higher rank. An isolated finite presentation certificate identifies the exact source base; the uniform conclusions are proved in writing. We also give an explicit counterexample to the source's thin-base auxiliary lemma and recover its faithful-base canonical-double-cover result with the repaired common theorem.

1 Two modules over a polygon group

Let $E = \mathbb{F}_2^2$, with basis $e = (1, 0)$ and $f = (0, 1)$, and put

$$B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

They are involutions, $Ce = e$, $Bf = f$, and BC has order 3. On the three nonzero vectors they generate the full symmetric group. For a positive integer m, let

$$D_m = \langle b, c \mid b^2 = c^2 = (bc)^{3m} = 1 \rangle$$

be the dihedral group of order $6m$. Its action on E sends b to B and c to C, and factors through the quotient $D_m \to D_1 \cong S_3$. Use this action on both summands of $V = E_0 \oplus E_3$, and form $G_m = V \rtimes D_m$. Multiplication is $(v, d)(w, h) = (v + d(w), dh)$. Set

$$s_0 = (e, 0; 1), \quad s_1 = (0, 0; b), \quad s_2 = (0, 0; c), \quad s_3 = (0, f; 1).$$

Each mark is an involution. They generate G_m: s_1, s_2 generate D_m, while their conjugates of s_0 and s_3 span the two independent copies of E. Nonconsecutive marks commute, because $Ce = e$, $Bf = f$, and translations commute. Moreover

$$|s_0 s_1| = 4, \quad |s_1 s_2| = 3m, \quad |s_2 s_3| = 4. \tag{1}$$

For example $(s_0 s_1)^2 = (e + Be, 0; 1) = (f, 0; 1) \neq 1$; the other outer calculation is dual.

Lemma 1. *The indicated marking of G_m is a string C-group. Its order is $96m$, its endpoint parabolics are $E_0 \rtimes D_m$ and $E_3 \rtimes D_m$, and its face numbers are $(4, 6m, 6m, 4)$.*

Proof. The two endpoint descriptions follow from the spanning argument above. Their intersection is precisely the common complement D_m, because $E_0 \cap E_3 = 0$. This is a direct application of the common criterion using image and kernel: their images in D_m coincide, and their translation kernels are disjoint.

Inside the first endpoint, $\langle s_0, s_1\rangle = E_0 \rtimes \langle b\rangle$ has order 8, and $\langle s_1, s_2\rangle = D_m$. Their intersection is exactly $\langle b\rangle$. Both are dihedral rank-two C-groups, so the rank-three endpoint has the intersection property. To make the inference precise, apply the recursive path criterion from *Intersections in simultaneous group actions*, Lemma 2, to its regular action: both polygon ends are polytopal and their intersection is the common marked subgroup. The dual argument proves the other endpoint, using $\langle s_2, s_3\rangle = E_3 \rtimes \langle c\rangle$ and intersection $\langle c\rangle$ with D_m. The same recursive criterion in rank 4, together with the endpoint intersection D_m, proves the full C property, including every subset of the four marks. It is not inferred just from adjacent orders.

The endpoint orders are $24m$, and $|G_m| = 16(6m) = 96m$. Deleting color 1 leaves $\langle s_0\rangle \times \langle s_2, s_3\rangle \cong C_2 \times D_8$, of order 16; deleting color 2 is dual. Indices give the four displayed face counts. □

Lemma 2. *Both endpoint flag graphs are nonbipartite, for every m.*

Proof. The three nonzero vectors in E_0 sum to zero. Conjugating s_0 by suitable words in b, c realizes these three translations. Their product is the identity. Written in the four marked involutions, this relation has odd length: each conjugated translation contributes one occurrence of s_0 and twice the length of its conjugating word. It is an odd closed walk in the first endpoint. The same argument in E_3 proves the second. This proves nonbipartiteness for all m; it does not depend on an order calculation in sample covers. □

2 The exact voltage construction

Write P_m for the regular polytope just proved. First consider $m = 1$. Its endpoint groups $E \rtimes S_3$ act faithfully as all 24 affine permutations of the four-element set E, and hence are S_4. The products $s_0s_1s_2$ and $s_1s_2s_3$ have order 3: the projected rotation has order 3 and its norm on E is $1+BC+(BC)^2 = 0$. Thus the facet and vertex-figure types are respectively $\{4, 3\}_3$ and $\{3, 4\}_3$. The exact universal presentation with these relations has order 96, as already certified in the accepted common projective marking paper's sixth universal row. The complete isolated program `base_certificate.g` verifies that exact presentation and its marked isomorphism to this 16-point affine permutation model; it is a required finite premise for this source identification, and its original attribution is retained. Since the displayed G_1 has order 96 and is a marked quotient of that presentation, this identifies it with the source base $\mathcal{B} = \{\{4, 3\}_3, \{3, 4\}_3\}$.

Use right multiplication for the colored Cayley adjacencies of G_1. Let $V = E_0 \oplus E_3$ denote its translation subgroup. Assign integer voltages to color 2 darts by

$$z(g, gs_2) = \begin{cases} 1 & g \in V, \\ -1 & g \in Vs_2, \\ 0 & \text{otherwise,} \end{cases}$$

and voltage0 to all other colors. The assignments on reversed darts are negatives, since $s_2^2 = 1$. Reduce modulo m for the cyclic cover.

Proposition 3. *This is, up to reversing all voltages and choosing the opposite edge pair, the voltage assignment of Leemans–Toledo Section 5. Its cover is the Cayley maniplex of G_m with the four marks above.*

Proof. An edge face of P_1 through the identity is $J = \langle s_0, s_2, s_3\rangle$, of order 16. Its intersection with V has order 8, and $VJ = V \rtimes \langle c\rangle$ has order 32. Hence VJ is the union of two edge faces. Vertices of P_1 identify with the four vectors of E_0; the endpoints of J are $0, e$, and those of the other edge face are the other two vectors. These are exactly an opposite edge pair in the source's K_4 skeleton.

The 16 origins in V are closed under colors 0 and3, and their color 1 and color 2 neighbors lie outside V. They are also closed under the word101, since $s_1s_0s_1$ is a translation. Thus they satisfy all

three orientation rules accompanying source Figure 5.1. These rules determine the orientation once one origin is selected. Indeed the moves $s_0, s_3, s_1s_0s_1$ preserve the sign, and s_2 reverses it. These moves generate VJ: the first and third span E_0, while s_3 and its conjugate by s_2 span E_3. Thus the 32 flags of the opposite edge pair have their signs uniquely determined. The two choices are V and Vs_2, as asserted.

The quotient of the base flag graph by V is the Cayley six-cycle of D_1 in colors 1,2, with semiedges in colors 0,3. Our voltage is pulled back from a voltage on that six-cycle: one directed color 2 edge has voltage1 and the other five edges have voltage0. Its derived cover is the alternating cycle of length $6m$, the Cayley graph of D_m in b, c. Its projection to the six-cycle is the marked quotient $D_m \to D_1$.

Pulling this cover back to the base gives the pairs

$$\{(g, d) \in G_1 \times D_m \,:\, gV = \overline{d} \in D_1\}.$$

The explicit map $(v, d) \mapsto ((v, \overline{d}), d)$ is a group isomorphism from $V \rtimes D_m$ onto this fiber product, with inverse $((v, \overline{d}), d) \mapsto (v, d)$. It sends the four displayed marks to the four colored adjacency pairs. The subgroup generated by them is the entire product, since it contains the complement D_m and both translation summands. Thus its Cayley graph is precisely the connected voltage cover, rather than merely one unidentified component. This proves the assertion for every modulus simultaneously. □

The kernel of $G_m \to G_1$ is $\langle (bc)^3 \rangle$, of order m. It is contained in both endpoint parabolics. Their full preimages are therefore connected endpoint faces. Each middle parabolic has order16 and maps isomorphically to the corresponding order16 base parabolic; its intersection with the kernel is trivial. Every base middle face therefore has m isomorphic lifts. This proves exactly the face-preimage clauses of source Lemma 5.1, rather than just their eventual consequences.

The source proves regularity by lifting four base automorphisms, including a piecewise lift with nine adjacency cases. The actual fiber product description makes those cases unnecessary: left multiplication in G_m is already regular on its Cayley graph. Lemma 1 also proves every intersection needed for polytopality directly. The retained source construction and its attribution are unchanged; this is a different complete derivation of that construction's properties.

For completeness, the source's displayed exceptional lift is also recovered. Choose the identity base flag in V, so ρ_i is left multiplication by s_i. Left multiplication by s_0, s_3 preserves the voltage, and left multiplication by s_2 negates it. For ρ_1, put $h(q) = 0$ on $q \in \{1, b\} \subset D_1$ and $h(q) = 1$ on the other four quotient flags. The lift is

$$(g, j) \longmapsto (s_1 g, -j + h(gV)). \tag{2}$$

The following complete six-state table verifies the only nonzero-voltage color. Its entries are integer equalities, so apply modulo every m.

q	1	b	c	bc	cb	bcb
$z(q, qc)$	1	0	−1	0	0	0
$z(bq, bqc)$	0	1	0	−1	0	0
$h(qc) - h(q)$	1	1	−1	−1	0	0

The second row is the negative of the first plus the third. Colors 0,3 do not change q, and right multiplication by b preserves h, so the required identity for all other colors is zero on both sides. The common lift calculation proves (2). Since $V \cup s_1V$ is exactly the source set $F_X \cup F_X^{\rho_1}$, this is its literal piecewise formula. A deck translation adjusts any of the other lifted base generators to the prescribed adjacent flag. Thus the lift statement and its concrete formula are retained, even though they are unnecessary as separate premises of the regularity proof.

3 The nonpolytopal family and all higher ranks

Theorem 4. *For each positive integer m, the canonical double cover of P_m is a finite regular faithful thin nonpolytopal rank-four maniplex with $192m$ flags and face numbers $(4, 12m, 12m, 4)$.*

Proof. Lemmas 1 and 2 verify every hypothesis of the common canonical-double-cover corollary in *Intersections and faces in graph covers*. Its proof gives all conclusions. In particular the vertex–facet intersections split into two components; faithfulness and thinness do not imply the missing strong connectivity. Distinct values of m give distinct flag counts and therefore nonisomorphic maniplexes. □

For clarity the failure also has a group description. Nonorientability makes the double-cover group the entire $G_m \times C_2$, with marks $(s_i, 1)$. Both endpoint parabolics contain the deck involution $(1, 1)$, because each has an odd relation. Their intersection is $D_m \times C_2$. The middle subgroup generated by $(b, 1), (c, 1)$ is the graph of the usual word-length parity homomorphism on the dihedral group, and has order $6m$. The deck involution is consequently an explicit element of the endpoint intersection outside the middle subgroup. This is the common fiber obstruction, not a failure detected only through a finite graph search.

Let M be any rank-r maniplex. Define $R(M)$ on $\mathrm{Flags}(M) \times \{0, 1\}$ by retaining all old adjacencies in each copy and adjoining color r which interchanges the copies. Its old i-faces are $F_i \times \{0, 1\}$ and its two new facets are $M \times \{0\}$ and $M \times \{1\}$. These descriptions show directly that faithfulness and thinness hold exactly when they do in M: maximal chains choose one new facet and then one old maximal chain; all old diamond conditions are unchanged, and the new top diamond has exactly the two new facets. A face-chain intersection either fixes one copy, or is the Cartesian product of its old intersection with a single edge. It is connected exactly when the old intersection is. Thus the connected-intersection property is preserved and reflected. For the faithful maniplexes used here it is equivalent to polytopality. The face description also proves the source's poset-polytopality equivalence without a faithfulness restriction. Maximal chains are pairs consisting of an old chain and one new facet. Within one new facet, strong connectivity is exactly that of the old poset. For chains in different new facets, first connect the old chains while fixing their common old faces, then change just the new facet. Conversely, applying strong connectivity to two chains in the same new facet forces a connection in that facet and hence in the old poset. Combined with the diamond calculation, this proves that the new face poset is a polytope exactly when the old one is. Every automorphism acts by the same old automorphism on both copies, optionally interchanging them, so $\mathrm{Aut}(R(M)) = \mathrm{Aut}(M) \times C_2$. Iterating this construction $r - 4$ times in Theorem 4 gives $192m\,2^{r-4}$ flags and proves the complete principal assertion of Leemans–Toledo Theorem 1.1 for every $r \geq 4$.

4 A thin base for which the auxiliary lemma fails

The general thin-base Lemma 3.5 in the source is false. All coordinates in the following example belong to $\mathbb{F}_2$. Start with the32 flags (v_0, v_1, v_2, v_3, z). The four adjacencies flip their indicated v_i; in addition color 0 adds v_1 to z and color 2 adds v_3 to z. Colors 1,3 leave z fixed. These are distinct fixed-point-free involutions, and nonconsecutive pairs commute. The graph is connected: the v_i can be changed freely, and the commutator of colors 0,1 changes z without changing them. The same is true of the commutator of colors 2,3.

Its i-faces are exactly the sets with fixed v_i. Indeed the remaining three colors change the other three v coordinates, and at least one of those two commutators remains available to change z. Thus every face has16 flags. Every pair of faces of distinct ranks meets, so the face poset has two elements at each proper rank, all comparable across different ranks. It is thin. It is unfaithful: fixing all four faces leaves both possible values of z.

Take a twofold voltage cover by adjoining $\epsilon \in \mathbb{F}_2$. Colors 0,3 have voltage0, color 1 has voltage

v_2, and color 2 has voltage v_3. Equivalently the four permutations are

color	coordinate flipped	z increment	ϵ increment
0	v_0	v_1	0
1	v_1	0	v_2
2	v_2	v_3	v_3
3	v_3	0	0

(3)

Each voltage is unchanged on reversing its edge. Directly from (3) the maps are distinct fixed-point-free involutions and the distant pairs commute. A color-1, 2 commutator changes only ϵ; the color-2, 3 commutator changes both z and ϵ. Hence the preimage of each 0-face is connected. Dually colors 0,1 change only z and colors 1,2 change only ϵ, so each 3-face preimage is connected. These facts also show the entire cover connected.

On a 1-face the voltage is the difference of the potential $z + v_0v_1$: this holds along each of its allowed colors 0,2,3 by (3). Its two lifts are therefore indexed by

$$(v_1, \alpha), \qquad \alpha = \epsilon + z + v_0v_1.$$

They each map isomorphically to that base face, by the elementary voltage-path argument. On a 2-face the potential is v_1v_2, and its two isomorphic lifts are indexed by

$$(v_2, \beta), \qquad \beta = \epsilon + v_1v_2.$$

All face-preimage hypotheses of the printed lemma are now verified. The cover has face counts $(2, 4, 4, 2)$ and is faithful: its faces determine all four v_i, then ϵ from β, and z from α.

Nevertheless it is not thin. Fix the 0-face $v_0 = 0$ and the 2-face $(v_2, \beta) = (0, 0)$. For every $(a, d) \in \mathbb{F}_2^2$, the flag

$$(v_0, v_1, v_2, v_3, z, \epsilon) = (0, a, 0, 0, d, 0)$$

lies in both these faces and in the 1-face $(v_1, \alpha) = (a, d)$. There are four intervening 1-faces, rather than two. This is a direct written counterexample to the exact thin-base statement. The independent finite diagnostic verifies all faces and diamond intervals, but is not a premise of this argument. The source proof's inference that an intersection of preimages has precisely the same component count as one preimage is not valid without a connectivity argument. The common theorem for faithful thin bases supplies that missing argument and proves all canonical-double-cover conclusions on their actual faithful hypothesis.

5 Attribution and the remaining source boundary

We use Leemans and Toledo's *Faithful and thin non-polytopal maniplexes*.

The fixed edition is arXiv:2305.01372v2, dated 10 May 2023, with 12 physical pages. Its main finite family, the voltage choice, the small-rank obstruction and the raviolo construction are attributed to that source; no historical novelty is asserted for our alternative derivation. The common paper includes the complete rank-three argument, the general faithful middle-face lemma and every clause of Lemma 3.8. Proposition 5.4 correctly prints "without the CIP", exactly as proved here.

The 11-cell and 57-cell examples follow from the same common corollary. Their retained local types are hemi-icosahedral facets with hemi-dodecahedral vertex-figures for the 11-cell, and the dual pair for the 57-cell. These projective maps are nonorientable, so their flag graphs have odd closed walks. The existence and regular polytopality of these objects of Grünbaum and Coxeter are the already accepted proper-end classifications in the projective-marking library; they are retained inputs for these illustrations, not new classifications here. Their double covers are consequently regular, faithful, thin and nonpolytopal. The informal exhaustive claim on source page 7 that the 192-flag example is the

smallest regular example is a separate ancillary census and is not reproduced. Nor do we re-establish the closing atlas counts of regular 11-cell covers, construct the proposed further voltage families, or prove the omitted four-facet extension statement.

The source's broad background assertion that Pos and Man are inverse for every thin flagged poset is not used. Our chain and face arguments start with maniplexes, and our inverse identifications with polytope flag graphs have their faithful and thin hypotheses explicitly proved. The source's graph-lifting attribution is retained in the common sufficient lift lemma. The cyclic cover here is an actual fiber product with a proved inverse map and proved generation; this is where the common theory replaces repeated local intersection and lift arguments. The finite base-presentation certificate, the written uniform proof, and bounded validation programs remain separate obligations. Four new Lean declarations prove only the generic equivariant-projection intersection criterion and the actual voltage-lift bijection and adjacency calculation. Two existing declarations concern simultaneous action fibers. Neither file formalizes face connectivity, the finite presentation calculation, or the uniform family; those application obligations are the written arguments above.